

# Assignment 9: Time Series Analysis

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Spring 2026

## OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

## Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

## Set up

1. Set up your session:
  - Check your working directory
  - Load the tidyverse, zoo, and trend packages
  - Set your ggplot theme

```
getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(zoo)
```

```
##  
## Attaching package: 'zoo'  
##  
## The following objects are masked from 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
library(trend)
```

```
mytheme <- theme_classic(base_size = 14) +  
  theme(  
    axis.text = element_text(color = "black"),  
    legend.position = "top"  
  )  
  
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone\_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1  
ozone_files <- list.files(  
  path = "Data/Raw/Ozone_TimeSeries",  
  pattern = "\\*.csv$",  
  full.names = TRUE  
)  
  
ozone_list <- lapply(ozone_files, read.csv)  
GaringerOzone <- bind_rows(ozone_list)  
  
dim(GaringerOzone)
```

```
## [1] 3589    20
```

## Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY\_AQI\_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```

# 3
GaringerOzone <- GaringerOzone %>%
  mutate(Date = as.Date(Date, format = "%m/%d/%Y"))

# 4
GaringerOzone <- GaringerOzone %>%
  select(Date,
         Daily.Max.8.hour.Ozone.Concentration,
         DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq(
  from = as.Date("2010-01-01"),
  to = as.Date("2019-12-31"),
  by = "day"
))

colnames(Days) <- "Date"

# 6
GaringerOzone <- left_join(Days, GaringerOzone, by = "Date")

dim(GaringerOzone)

```

```
## [1] 3652    3
```

```
sum(is.na(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration))
```

```
## [1] 63
```

```
head(GaringerOzone)
```

```
##           Date Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
## 1 2010-01-01                      0.031                      29
## 2 2010-01-02                      0.033                      31
## 3 2010-01-03                      0.035                      32
## 4 2010-01-04                      0.031                      29
## 5 2010-01-05                      0.027                      25
## 6 2010-01-06                      NA                       NA
```

## Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

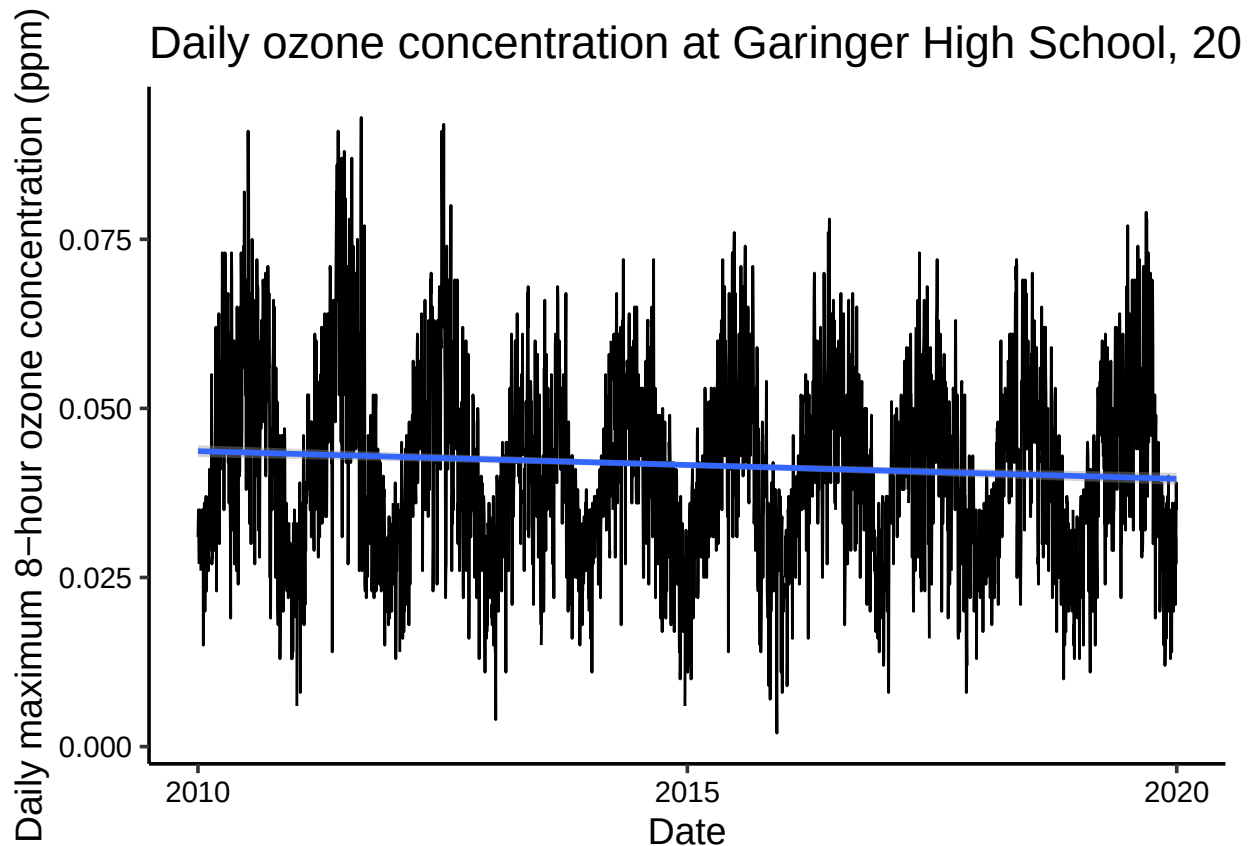
```

#7
ggplot(GaringerOzone,
  aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line(na.rm = TRUE) +

```

```
geom_smooth(method = "lm", se = TRUE, na.rm = TRUE) +
labs(
  x = "Date",
  y = "Daily maximum 8-hour ozone concentration (ppm)",
  title = "Daily ozone concentration at Garinger High School, 2010 to 2019"
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Answer: The plot suggests a slight downward trend in ozone concentration over time. Daily values vary a lot and there is a clear seasonal pattern, but the fitted linear trend line indicates that ozone concentrations generally decreased a little over the 2010s.

## Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone <- GaringerOzone %>%
  mutate(
```

```
Ozone_interp = na.approx(
  Daily.Max.8.hour.Ozone.Concentration,
  x = Date,
  na.rm = FALSE
)
)
```

Answer: We used linear interpolation because it fills missing values in a simple and reasonable way using nearby observations. A piecewise constant interpolation would create abrupt jumps, which is not realistic for ozone data. A spline interpolation would make the series too smooth and could introduce patterns that were not actually observed.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(
    Year = year(Date),
    Month = month(Date)
  ) %>%
  group_by(Year, Month) %>%
  summarise(
    Mean_Ozone = mean(Ozone_interp, na.rm = TRUE),
    .groups = "drop"
  )

GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(Date = as.Date(paste(Year, Month, "01", sep = "-")))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

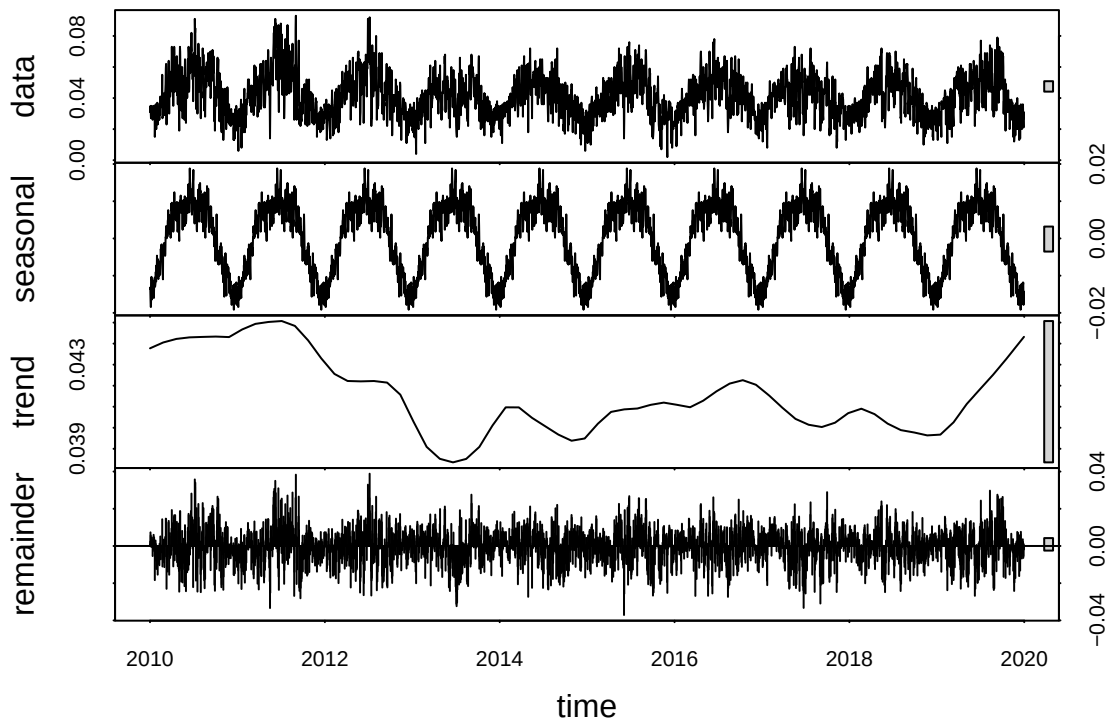
```
#10
GaringerOzone.daily.ts <- ts(
  GaringerOzone$Ozone_interp,
  start = c(2010, 1),
  end = c(2019, 365),
  frequency = 365
)

GaringerOzone.monthly.ts <- ts(
  GaringerOzone.monthly$Mean_Ozone,
  start = c(2010, 1),
  end = c(2019, 12),
  frequency = 12
)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

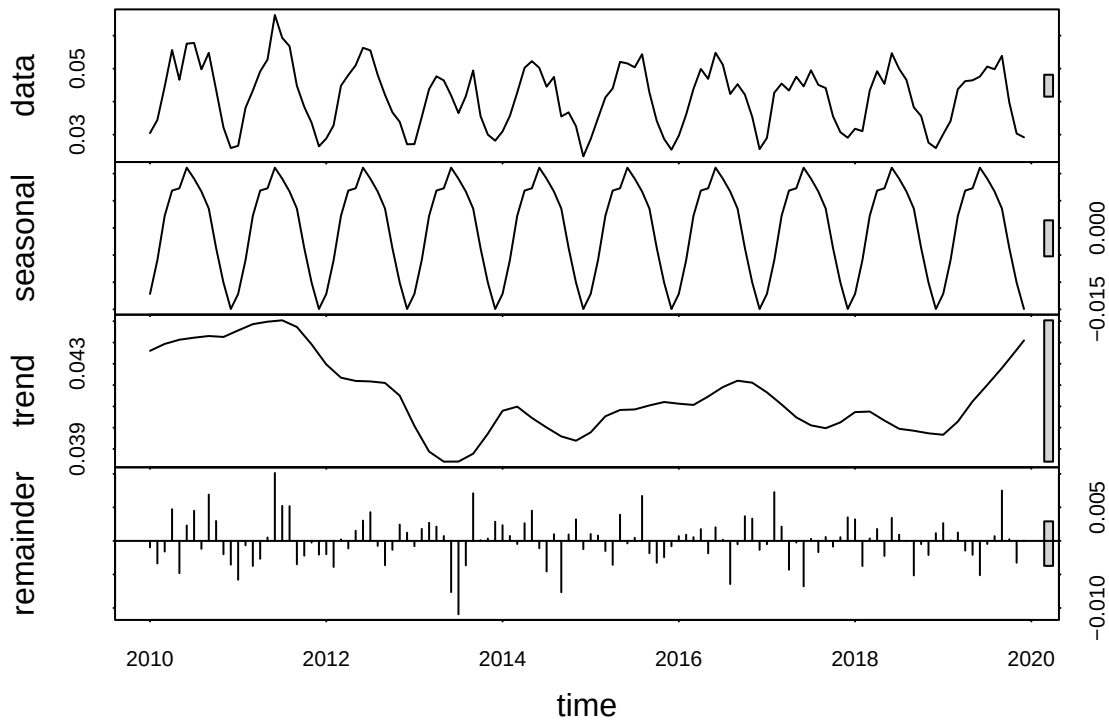
```
#11
# daily time series decomposition
GaringerOzone.daily.decomp <- stl(
  GaringerOzone.daily.ts,
  s.window = "periodic"
)

plot(GaringerOzone.daily.decomp)
```



```
# monthly time series decomposition
GaringerOzone.monthly.decomp <- stl(
  GaringerOzone.monthly.ts,
  s.window = "periodic"
)

plot(GaringerOzone.monthly.decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
Ozone_trend_test <- smk.test(GaringerOzone.monthly.ts)

Ozone_trend_test

##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
## S varS
## -77 1499

summary(Ozone_trend_test)

##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
```

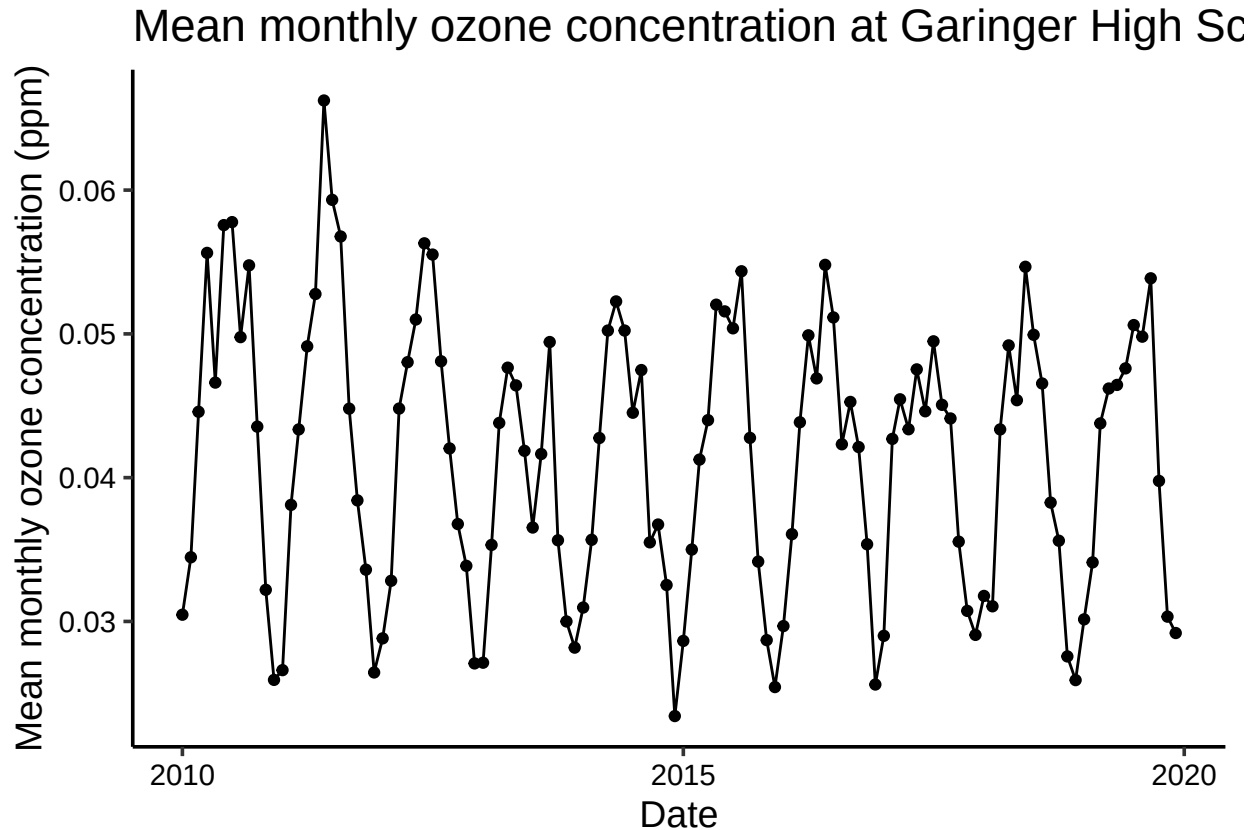
```
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
##      S varS    tau      z Pr(>|z|)
## Season 1:  S = 0   15  125  0.333  1.252  0.21050
## Season 2:  S = 0   -1  125 -0.022  0.000  1.00000
## Season 3:  S = 0   -4  124 -0.090 -0.269  0.78762
## Season 4:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 5:  S = 0  -15  125 -0.333 -1.252  0.21050
## Season 6:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 7:  S = 0  -11  125 -0.244 -0.894  0.37109
## Season 8:  S = 0   -7  125 -0.156 -0.537  0.59151
## Season 9:  S = 0   -5  125 -0.111 -0.358  0.72051
## Season 10: S = 0  -13  125 -0.289 -1.073  0.28313
## Season 11: S = 0  -13  125 -0.289 -1.073  0.28313
## Season 12: S = 0   11  125  0.244  0.894  0.37109
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer: The seasonal Mann-Kendall test is appropriate because ozone concentrations show strong seasonal patterns across months. Since ozone levels vary systematically by season, using a standard Mann-Kendall test could confound seasonal variation with long-term trends. The seasonal Mann-Kendall test accounts for these recurring patterns and provides a more accurate assessment of monotonic trends over time.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly,
       aes(x = Date, y = Mean_Ozone)) +
  geom_point() +
  geom_line() +
  labs(
    x = "Date",
    y = "Mean monthly ozone concentration (ppm)",
    title = "Mean monthly ozone concentration at Garinger High School, 2010 to 2019"
  )
```





14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Mean monthly ozone concentrations appear to decrease slightly over time, although there is strong seasonal variation. The overall pattern in the plot suggests a gradual decline. The Seasonal Mann-Kendall test supports this, indicating a statistically significant trend, though it is relatively weak ( $z = -1.963$ ,  $p = 0.04965$ ).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
GaringerOzone.monthly.nonseas <-
  GaringerOzone.monthly.ts -
  GaringerOzone.monthly.decomp$time.series[, "seasonal"]

#16
Ozone_nonseas_trend <- mk.test(GaringerOzone.monthly.nonseas)

Ozone_nonseas_trend
```

```
##
## Mann-Kendall trend test
##
## data:  GaringerOzone.monthly.nonseas
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## -1.179000e+03  1.943657e+05 -1.651376e-01
```

```
summary(Ozone_nonseas_trend)
```

```
##          Length Class  Mode
## data.name    1      -none- character
## p.value      1      -none- numeric
## statistic    1      -none- numeric
## null.value   1      -none- numeric
## parameter    1      -none- numeric
## estimates    3      -none- numeric
## alternative  1      -none- character
## method       1      -none- character
## pvalg        1      -none- numeric
```

Answer: After removing the seasonal component, the Mann-Kendall test shows a stronger decreasing trend in ozone concentrations ( $z = -2.672$ ,  $p = 0.00754$ ). This is consistent with the Seasonal Mann-Kendall test, which also indicated a negative trend, but the result is more statistically significant after removing seasonality. This suggests that seasonal variation was partially masking the underlying trend, and that there is a clear long-term decrease in ozone levels over time.